

1 Shallow-Water Model

Waves, rotation, instability and large-scale flow

1.1 Gravity waves and numerical stability

Prediction. For a layer depth of about 10 km, estimate the shallow-water gravity-wave speed $c = \sqrt{gh}$. Roughly how far should the wave travel in one hour?

Expected prediction. Taking $g = 9.81 \text{ m s}^{-2}$ and $h \simeq 10^4 \text{ m}$,

$$c = \sqrt{9.81 \times 10^4} \simeq 3.13 \times 10^2 \text{ m s}^{-1}.$$

In one hour the corresponding distance is

$$ct \simeq 313 \times 3600 \simeq 1.13 \times 10^6 \text{ m},$$

or about 1100 km. On an axis labelled in 1000 km, the wave should therefore move a little over one axis unit per hour.

Exercise 1. Estimate the propagation speed from `gravity.gif` and compare it with $\sqrt{gh}$. Is there much signal in the relative-vorticity panel, and why?

Model answer. The propagating disturbance moves at approximately $3 \times 10^2 \text{ m s}^{-1}$, consistent with the expected $\sqrt{gh}$ speed for a layer roughly 10 km deep. The relative-vorticity signal is weak. With $f = \beta = 0$ the initial height perturbation mainly excites a divergent pressure–gravity-wave response: air accelerates away from and towards height anomalies, but there is little rotational motion. Gravity waves therefore appear strongly in the height and wind field without requiring a large ζ signal.

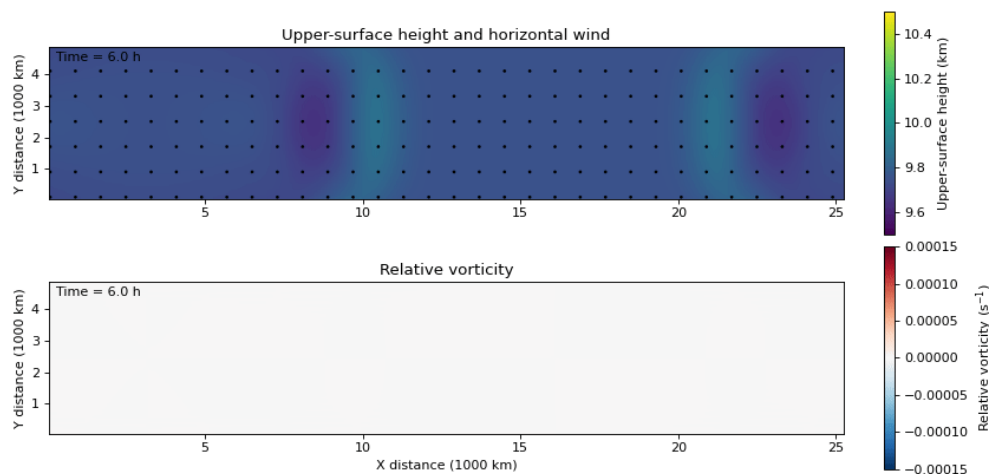


Figure 1: Representative frame from the gravity-wave calculation. The disturbance propagates rapidly while the relative-vorticity field remains weak.

Exercise 2. What changes physically and numerically when `dt_mins` is increased from 1 to 5 minutes? Why is the resulting failure not a real atmospheric or oceanic process?

Model answer. Nothing in the physical experiment has changed: gravity, fluid depth, initial height perturbation, domain and boundary conditions are the same. Only the numerical time-step has changed. The larger time-step makes the Courant number exceed the stability range of the explicit Lax–Wendroff calculation, so information effectively travels too far across the grid during one numerical update. The invalid fluid depths or rapidly growing values are therefore a numerical-stability failure, not a physical instability of the shallow-water system.

1.2 A tsunami analogue: changing fluid depth

Prediction. What should happen to wave speed and wavelength when the gravity wave enters the shallower water over the seamount?

Expected prediction. Since $c = \sqrt{gh}$, reducing h reduces the propagation speed. If the wave period remains approximately unchanged, $\lambda = cT$ also decreases, so the wavelength should shorten over the shallower region.

Exercise 3. Identify the changes as the wave crosses the shallower region. Which features resemble tsunami shoaling?

Model answer. The part of the wave over the seamount travels more slowly than the part remaining in deeper water. The wavefront therefore becomes distorted and compressed over the shallower region, with a shorter local wavelength and a tendency for the disturbance to sharpen or increase in amplitude. These are qualitative features of tsunami shoaling. The comparison is deliberately idealised: the seamount geometry and initial free-surface displacement are much simpler and larger than in a real tsunami.

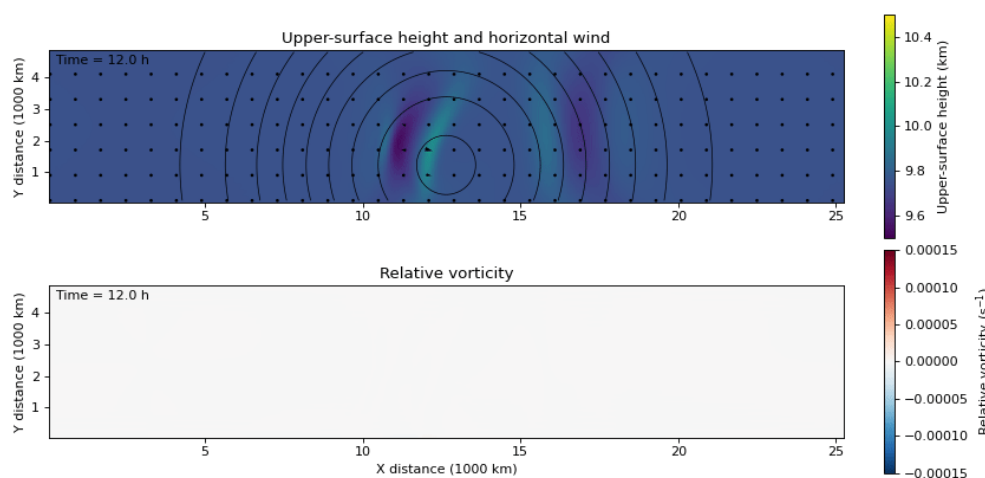


Figure 2: Representative frame from the tsunami analogue. The shallower region modifies the speed and shape of the propagating gravity wave.

1.3 Barotropic instability

Prediction. An instability amplifies perturbations, but does it create a perturbation from an exactly symmetric state? Which jet calculation should develop visible waves first?

Expected prediction. The run with `add_random_height_noise=True` should develop visible waves first. An unstable basic state amplifies an existing disturbance; in an exactly symmetric numerical state there may be no finite perturbation available to grow, apart from very small round-off or discretisation errors.

Exercise 4. Compare `jet_noise.gif` and `jet_no_noise.gif`. Why does the random perturbation matter even though the underlying jet is unstable?

Model answer. The perturbed jet develops clear meanders and vortical structures, whereas the unperturbed calculation remains much more nearly zonally symmetric over the same integration time. The jet can be dynamically unstable while still requiring a seed disturbance. The random height perturbation supplies that seed explicitly; the unstable modes then extract energy from the sheared basic flow and grow. The comparison separates the property of the basic state (unstable) from the finite disturbance whose growth makes the instability visible.

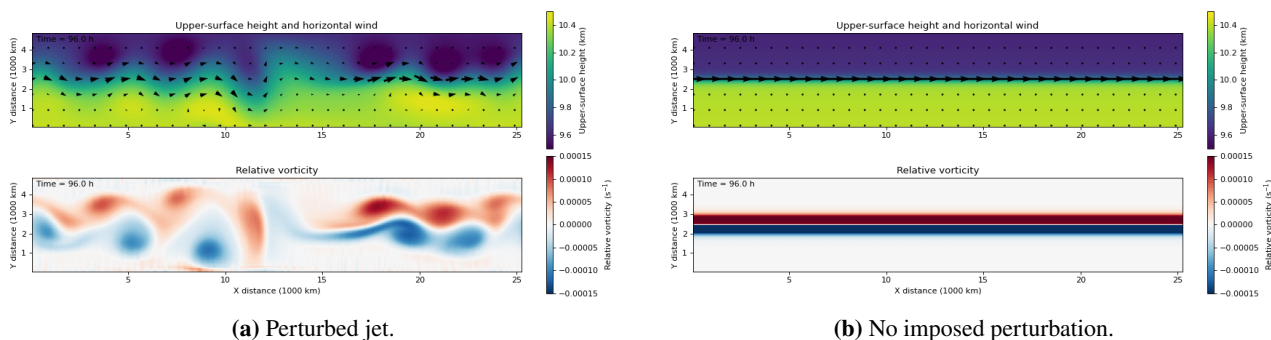


Figure 3: The random perturbation seeds the unstable mode and allows the meanders to become visible within the four-day integration.

Optional sharp-shear experiment

Model answer. The sharper shear layer is even more susceptible to disturbance growth. In the supplied animation the initially narrow shear zone rolls into pronounced waves and vortices, giving a more frontal or cyclogenesis-like appearance than the broader zonal jet. The important point is again that the growing structures draw energy from the horizontal shear; they are not imposed directly by the random perturbation.

1.4 Orographic Rossby waves

Exercise 5. Explain the initial meander in terms of potential-vorticity conservation as the flow passes over the Gaussian mountain. Where is relative vorticity generated, and why?

Model answer. For shallow-water flow the potential vorticity is approximately

$$q = \frac{f + \zeta}{h}.$$

As a column moves up the mountain, the fluid layer becomes shallower. If q is approximately conserved, the absolute vorticity $f + \zeta$ must decrease as h decreases. In the Northern-Hemisphere-like setup this produces an anticyclonic (negative- ζ) tendency over or just downstream of the rising topography. As the column descends and the layer deepens, $f + \zeta$ must increase, producing a cyclonic (positive- ζ) tendency. The alternating vorticity anomalies deflect the westerly flow and initiate the meander.

Exercise 6. Why do NaN values appear when $f = 0$ and $\beta = 0$ while `initially_geostrophic=True`? Is this the physical response of a non-rotating westerly?

Model answer. The failure occurs while the initial state is being constructed. The geostrophic-wind calculation contains f in the denominator, schematically

$$u_g \propto -\frac{g}{f} \frac{\partial h}{\partial y}, \quad v_g \propto \frac{g}{f} \frac{\partial h}{\partial x}.$$

With $f = 0$ everywhere these expressions are undefined, so division by zero produces infinities or NaN values before the dynamical experiment can begin. It is therefore not the physical evolution of a 20 m s^{-1} non-rotating westerly. It demonstrates that changing a physical parameter can invalidate an assumption used to initialise the model.

Exercise 7. Compare the f -plane and β -plane mountain calculations. Why is the downstream response different when f has no meridional gradient?

Model answer. Both calculations generate local vorticity as the flow is stretched and compressed over the mountain. On the β plane, however, a north–south displacement also changes the planetary vorticity experienced by a fluid column. Conservation of potential vorticity then provides the restoring mechanism for a Rossby wave, producing a clearer large-scale meander or downstream wave train. On the f plane there is no planetary-vorticity gradient, so that Rossby-wave restoring mechanism is absent; the topographically generated disturbance is correspondingly more local or simply advected downstream.

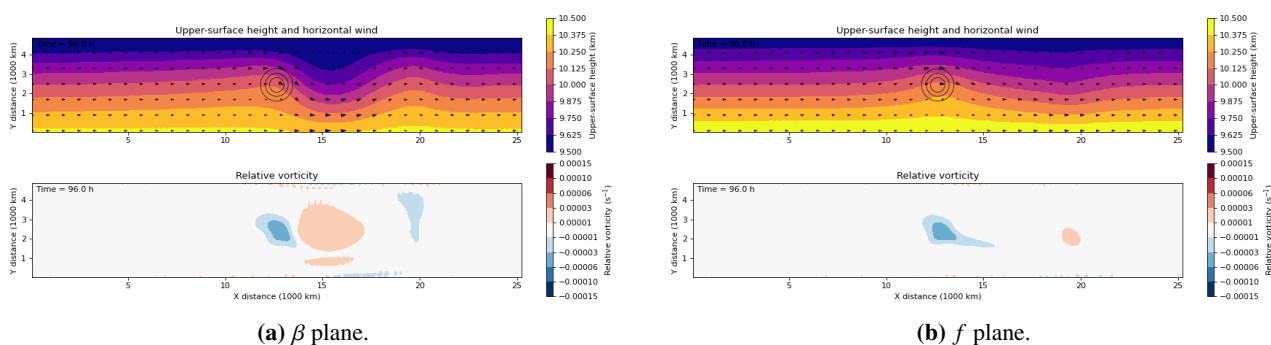


Figure 4: The meridional gradient of planetary vorticity changes the downstream response to the same mountain forcing.

1.5 Flow over real Earth orography

Prediction. Where should the largest disturbances to the uniform westerly flow occur: over flat ocean, over major topography, or downstream of it?

Expected prediction. The strongest response should be associated with major topography and its downstream wake. Orography changes the fluid-layer depth, forcing potential-vorticity changes and disturbing the geostrophically balanced westerly. Flat ocean regions provide much weaker direct forcing, although waves generated elsewhere can propagate across them.

Exercise 8. Compare the Earth-orography run with the Gaussian-mountain β -plane run. Which features are tied to major topography, which propagate downstream, and why is the response more complicated?

Model answer. Large disturbances are generated over and immediately downstream of major terrain. Some height and vorticity features remain approximately phase-locked to the topographic forcing, while other meanders extend and propagate downstream as large-scale waves. The response is more complicated than for the single Gaussian mountain because many mountain ranges and continental-scale height variations force the flow simultaneously. Their wave responses can overlap and interfere, and the real-orography field contains a wide range of horizontal scales rather than one isolated obstacle.

Exercise 9. Which important aspects of the real atmosphere are absent from this one-layer shallow-water experiment?

Model answer. The model has no explicit vertical atmospheric structure. It therefore cannot represent vertical stratification and shear, baroclinic instability, vertical propagation of waves, a realistic tropopause, boundary-layer turbulence, moist convection, clouds and latent heating, radiative processes, or detailed surface drag. The horizontal planetary-scale flow can look recognisable, but it should not be interpreted as a complete weather or climate model.

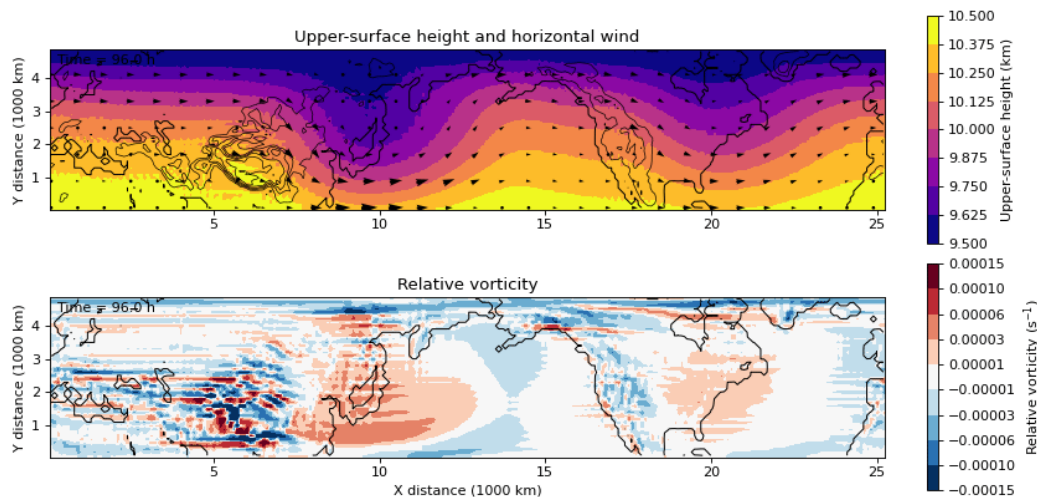


Figure 5: Final frame from the Earth-orography calculation. The most structured response is associated with major topography and its downstream influence.

1.6 Further experiments if time permits

Jupiter-like alternating jets

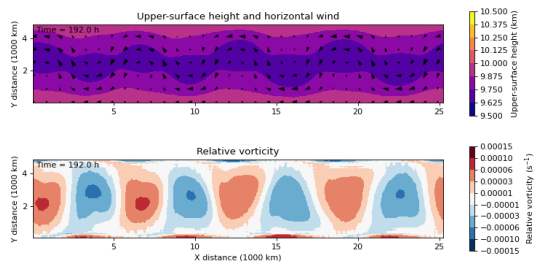
Model answer. The alternating zonal jets become wavy and generate coherent vortices. Vortices can interact and merge, leaving fewer, larger structures: this is qualitatively consistent with the upscale transfer of energy in two-dimensional or strongly geostrophic turbulence. The comparison with Jupiter's Great Red Spot is qualitative only; the shallow-water model omits much of Jupiter's vertical atmospheric structure and forcing.

Equatorially trapped waves

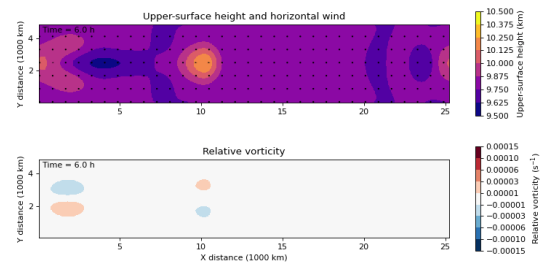
Model answer. Because $f = \beta y$ changes sign across the centre of the domain, the equatorial region acts as a waveguide. The supplied calculation develops disturbances concentrated near the model equator, with organised height and vorticity structures rather than a disturbance spreading uniformly to all latitudes. The trapping reflects the change in rotational dynamics with distance from the equator.

Equatorial Kelvin wave

Model answer. The eastward-propagating component remains concentrated close to the equator and travels at approximately the shallow-water gravity-wave speed $\sqrt{gh}$. The westward response is not its mirror image: it disperses into other equatorial wave components and is less cleanly trapped. Rotation therefore breaks the east–west symmetry that would exist for a non-rotating gravity wave. With h close to 10 km, $\sqrt{gh}$ is about 310 m s^{-1} , consistent with the motion of the eastward pulse in the supplied animation.



(a) Equatorially trapped disturbance.



(b) Kelvin-wave experiment.

Figure 6: Examples of equatorial trapping in the shallow-water model.